THE GYSIN MAP ON A RESTRICTED CLASS IS THE CUP PRODUCT OF THE PULLED BACK CLASS WITH THE GYSIN CLASS OF THE UNIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The global Gysin setup and the maps on local cohomology. Same chapter derham.tex, label lemma-gysin-global (cited at statement line 5521). It fixes the situation of $i : Z \to X$ and provides the maps $\gamma^p : \Omega^p_{Z/S} \to \mathcal{H}^c_Z(\Omega^{p+c}_{X/S})$ whose interaction with the cup product is the assertion.
- The cup product for local cohomology with support, and the general cup product formalism. Cohomology, remark cohomology-remark-support-cup-product (invoked at proof line 5551) and Section \ref{cohomology-section-cup-product} (invoked at proof line 5577). The first supplies the canonical maps from the tensor product with $R\mathcal{H}_Z(\Omega^c_{X/S})$ into the local cohomology of the tensor product, the second lets the claim be reduced to the commutativity of a single diagram.
- Vanishing of the local cohomology sheaves above the codimension. Derived Categories of Schemes, label perfect-lemma-supported-trivial-vanishing (invoked at proof line 5560). It shows that $R\mathcal{H}_Z(\Omega^j_{X/S})$ has no cohomology above degree c, so passing to degree c cohomology sheaves is legitimate.

Source reference. Original remark 0G75: sources/stacks-chow/tags.txt, source lines 20220–20220. Navigation pointer for \ref{cohomology-remark-support-cup-product} used inside the counted span (line 5551); the argument that consumes it is retained in the span.

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Source reference. Original section 0FKU: sources/stacks-chow/tags.txt, source lines 19474–19474. Navigation pointer for \ref{cohomology-section-cup-product} used inside the counted span (line 5577); the argument that consumes it is retained in the span.

Source reference. Original lemma 0G86: sources/stacks-derham/pinned-original-source.tex, source lines 5473–5473. Navigation pointer for \ref{lemma-gysin-global} used inside the counted span (line 5521); the argument that consumes it is retained in the span.

Source reference. Original lemma 0G7N: sources/stacks-chow/tags.txt, source lines 20238–20238. Navigation pointer for \ref{perfect-lemma-supported-trivial-vanishing} used inside the counted span (line 5561); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $X \to S$ and $i : Z \to X$ be as in Lemma 0G86. Given $\alpha \in H^q(X, \Omega^p_{X/S})$ we have $\gamma^p(\alpha|_Z) = i^{-1}\alpha \wedge \gamma^0(1)$ in $H^q(Z, \mathcal{H}^c_Z(\Omega^{p+c}_{X/S}))$. Please see proof for notation.*

Proof. The restriction $\alpha|_Z$ is the element of $H^q(Z, \Omega^p_{Z/S})$ given by functoriality for Hodge cohomology. Applying functoriality for cohomology using $\gamma^p : \Omega^p_{Z/S} \to \mathcal{H}^c_Z(\Omega^{p+c}_{X/S})$ we get get $\gamma^p(\alpha|_Z)$ in $H^q(Z, \mathcal{H}^c_Z(\Omega^{p+c}_{X/S}))$. This explains the left hand side of the formula.

To explain the right hand side, we first pullback by the map of ringed spaces $i : (Z, i^{-1}\mathcal{O}_X) \to (X, \mathcal{O}_X)$ to get the element $i^{-1}\alpha \in H^q(Z, i^{-1}\Omega^p_{X/S})$. Let $\gamma^0(1) \in H^0(Z, \mathcal{H}^c_Z(\Omega^c_{X/S}))$ be the image of $1 \in H^0(Z, \mathcal{O}_Z) = H^0(Z, \Omega^0_{Z/S})$ by γ^0. Using cup product we obtain an element

$$i^{-1}\alpha \cup \gamma^0(1) \in H^{q+c}(Z, i^{-1}\Omega^p_{X/S} \otimes_{i^{-1}\mathcal{O}_X} \mathcal{H}^c_Z(\Omega^c_{X/S}))$$

Using Cohomology, Remark 0G75 and wedge product there are canonical maps

$$i^{-1}\Omega^p_{X/S} \otimes^{\mathbf{L}}_{i^{-1}\mathcal{O}_X} R\mathcal{H}_Z(\Omega^c_{X/S}) \to R\mathcal{H}_Z(\Omega^p_{X/S} \otimes^{\mathbf{L}}_{\mathcal{O}_X} \Omega^c_{X/S}) \to R\mathcal{H}_Z(\Omega^{p+c}_{X/S})$$

By Derived Categories of Schemes, Lemma 0G7N the objects $R\mathcal{H}_Z(\Omega^j_{X/S})$ have vanishing cohomology sheaves in degrees $> c$. Hence on cohomology sheaves in degree c we obtain a map

$$i^{-1}\Omega^p_{X/S} \otimes_{i^{-1}\mathcal{O}_X} \mathcal{H}^c_Z(\Omega^c_{X/S}) \longrightarrow \mathcal{H}^c_Z(\Omega^{p+c}_{X/S})$$

The expression $i^{-1}\alpha \wedge \gamma^0(1)$ is the image of the cup product $i^{-1}\alpha \cup \gamma^0(1)$ by the functoriality of cohomology.

Having explained the content of the formula in this manner, by general properties of cup products (Cohomology, Section 0FKU), it now suffices to prove that the diagram

$$\begin{array}{ccccc} i^{-1}\Omega^p_X \otimes \Omega^0_Z & & \xrightarrow[\mathrm{id}\otimes\gamma^0]{} & & i^{-1}\Omega^p_X \otimes \mathcal{H}^c_Z(\Omega^c_X) \\ \downarrow & & & & \downarrow \wedge \\ \Omega^p_Z \otimes \Omega^0_Z & \xrightarrow{\wedge} & \Omega^p_Z & \xrightarrow{\gamma^p} & \mathcal{H}^c_Z(\Omega^{p+c}_X) \end{array}$$

is commutative in the category of sheaves on Z (with obvious abuse of notation). This boils down to a simple computation for the maps $\gamma^j_{f_1,\ldots,f_c}$ which we omit; in fact these maps are chosen exactly such that this works and such that 1 maps to $\frac{\mathrm{d}f_1 \wedge \ldots \wedge \mathrm{d}f_c}{f_1 \ldots f_c}$. □